

Problem 1. Use the table below to compute numerically $\lim_{x \rightarrow 0} \frac{e^{3x}-1}{x}$:

x	-0.1	-0.01	-0.001	0	0.001	0.01	0.1
$f(x)$				—			

Problem 2. Evaluate:

$$\lim_{x \rightarrow 0} \frac{\sin(x) - x}{4x}$$

Problem 3. *Locate and classify all discontinuities of the function $f(x)$ below:*

$$f(x) = \begin{cases} -1 & x < -1 \\ \frac{x^2+x}{2x} & -1 \leq x \leq 1 \\ -\cos(\pi x) & x > 1 \end{cases}$$

Problem 4. Use the definition of the derivative to compute $f'(x)$ if $f(x) = x^2 - x$.

Problem 5. Find the equation of the line tangent to $f(x) = \sqrt{x}$ at $x = 4$

Problem 6. *Differentiate the following functions:*

(i) $f(x) = 3x^4 + 7\sqrt[4]{x^3} - 5x^{-\frac{2}{5}} + 9x + 1$

(ii) $g(x) = e^x \csc(x)$

(iii) $h(x) = \frac{\sec(x)}{\sin(x)}$

(iv) $s(x) = \tan(x^2 + 3x)$

Problem 7. *The position of a dampened pendulum is measured from the angle (in radians) from its resting point. Its position after t seconds is given by the equation:*

$$s(t) = e^{-t} \cos\left(\frac{\pi}{2}t\right)$$

(i) *What is the velocity function $v(t)$ (in radians per second)?*

(ii) *What is the velocity after 1 second?*

Problem 8. *The limit below represents the derivative of some function $f(x)$ at a point a :*

$$\lim_{h \rightarrow 0} \frac{\sqrt[3]{\frac{1}{64+h}} - \frac{1}{4}}{h}$$

(i) *Find the function $f(x)$ and the point a .*

(ii) *Compute the limit.*